

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® Service Tools required.

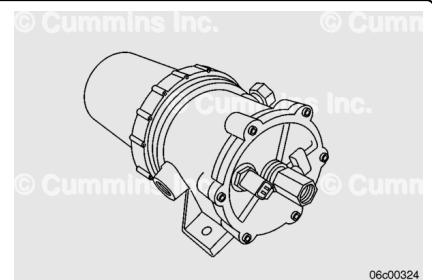
Additional Service Items

- No additional service items required.

General Information

The fuel system requires the use of two fuel filters. The suction side filter **must** have the following characteristics:

- Water-separating
- 10 micron rating
- Water in fuel sensor with shunt resistor
- Water drain valve
- Engine or chassis mounted.



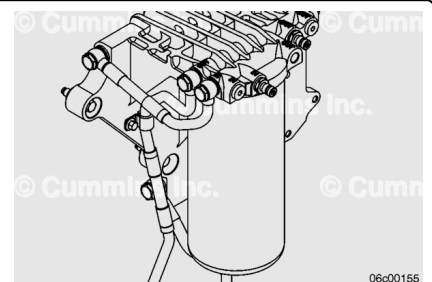
The pressure-side filter **must** have the following characteristics:

- 4 micron rating
- Engine mounted.

Use the following procedure for fuel filter recommendations.

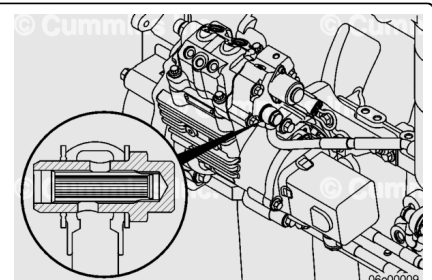
Refer to Procedure 018-024 in Section V.

(</qs3/pubsys2/xml/en/procedures/377/377-018-024.html>)



The system also contains a fuel screen located in the banjo bolt at the high-pressure fuel pump inlet.

- Rating
 - 86 micron
- Location



- inside the banjo bolt
- Service
 - replace the banjo bolt

Preparatory Steps

⚠ WARNING ⚠

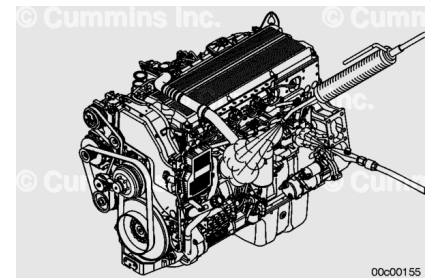
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.), which would expose the fuel system or internal engine components to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 3.

(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

Clean the fuel filter head with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

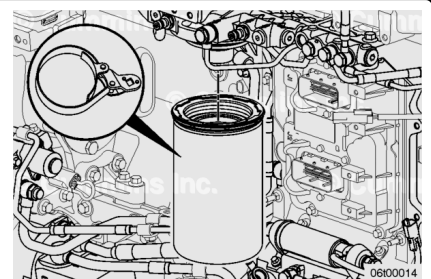
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

Remove

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Remove the fuel filter with filter wrench, Part Number 3400157.

Make sure the seal ring does **not** stick to the filter head.

Remove the ring with an o-ring pick, if necessary.

Install

Use the correct filter(s) for the engine. Filters **must** remove a minimum of 95 percent of free and emulsified water. They **must** also have a 4-micron particle-removal efficiency.

Refer to Procedure 018-024 in Section V.

(</qs3/pubsys2/xml/en/procedures/377/377-018-024.html>)

Apply a thin coating of clean engine oil to the filter gasket surface.

Note : Cummins® Fleetguard® filters come equipped with a disposable insert that will prevent fuel from

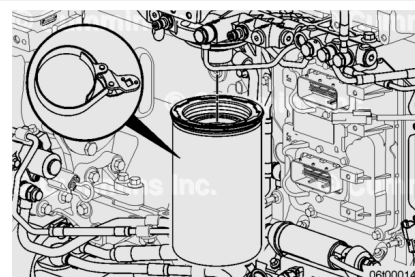


entering the clean side of the filter media during prefilling.

Note : If the engine is not equipped with a lift pump, the fuel filter will have to be properly prefilled before installation. Refer to Procedure 005-234 in Section A. (</qs3/pubsys2/xml/en/procedures/377/377-005-234-om.html>)

⚠ CAUTION ⚠

Mechanical overtightening of the filter can distort the threads or damage the filter element seal.



Note : Engines equipped with priming pumps do not require the fuel filter to be filled prior to installation, if the following filling procedure is followed.

Install the filter onto the filter head. Turn the filter until the gasket contacts the filter head surface.

Tighten the filter an additional $\frac{3}{4}$ of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Prime the fuel system. Refer to Procedure 005-234 in Section A. (</qs3/pubsys2/xml/en/procedures/377/377-005-234-om.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks and proper operation.

